[INCH-POUND]
A-A-3037
October 10, 1996

COMMERCIAL ITEM DESCRIPTION

LIFTER, VALVE SPRING

The General Services Administration has authorized the use of this Commercial Item Description for procurement of type I, class 1, style A lifter, specified in GGG-L-350 for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers a valve spring lifter used for removing and replacing the valve springs of gasoline engines.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The lifter shall consist essentially of a C-shaped frame, a rack and pinion gear or screw, or actuating lever or levers, and an operating handle and jaws. The lifters shall be suitable for removing and replacing valve springs of L-head, V, and overhead valve engines. The frame, jaws, rack and pinion gear, or screws shall be of steel. Other components shall be made of steel or other materials suitable for the purpose intended. All moving parts of the lifter shall operate freely without bending or binding during spring removing or replacing operations. The lifter shall be provided with a means for setting and maintaining the required distance between the jaws and the end of the rack screw. When a setting is established, subsequent springs shall be lifted by actuating the operating handle. All parts of the same manufacturer which may be subject to replacement during the normal life of the tool shall be completely interchangeable and can be replaced or adjusted without requiring modification. All metal components not of corrosion-resistant material shall be coated with zinc plate, or chemically produced oxide. The lifter shall be similar to Figure 1 and conform to the dimensions specified in Table I.

3.1.1 Jaws. The lifter shall be provided with one set of straight jaws, one set of jaws suitable for General Motors and Continental engines, and one set of special offset jaws. Additional jaws may be provided when required (see 7.2). The jaws shall be flanged or provided with a means that will retain the spring in a normal position on the jaws. The jaws shall be heat-treated to between 33-43 HRC.

3.2 Test load. Each sample lifter shall have a test load of 450 pounds minimum applied between the jaws and the end of the rack or screw in line with the axis of the direction of motion (axis of screw or rack), the direction to be such as to cause the lifter to open. After removal of the test load there shall be no evidence of permanent deformation or damage to any part of the lifter. Failure to meet this test shall be cause for rejection of the lifter and the lot it represents.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

Distribution Statement A. Approved for public release; distribution is unlimited.

FSC 5120

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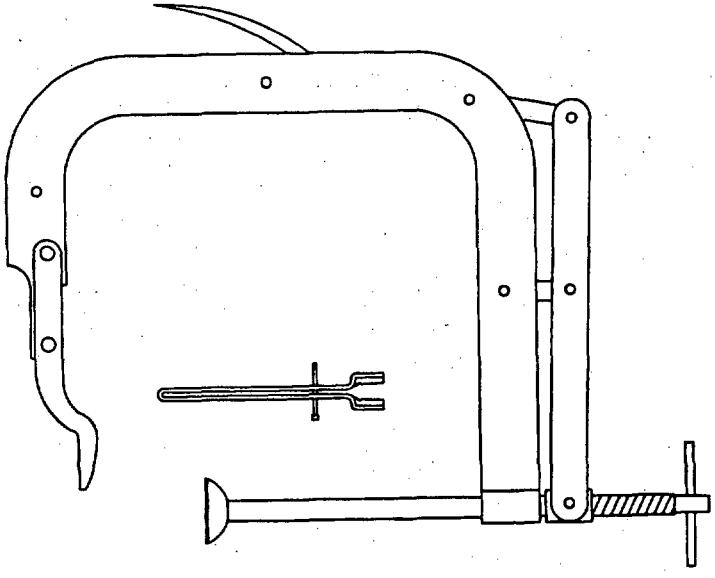


Figure 1. Lifter, C-frame, with depth adjustment.

TABLE I. Lifter, C-frame, with depth adjustment

Capacity (valve spring outside diameter)	A		B (full open position)		Weight (approx.)	Test load (minimum)
(Inches)	Min. (Inches)	Max. (Inches)	Min. (Inches)	Max. (Inches)	(Pounds)	(Pounds)
7/8 to 1-1/2	9-3/4	10-1/4	10	10-1/2	7-1/4	450

3.3 Hardness test. Sample lifter jaws shall be tested in accordance with ASTM E18 to determine conformance to a hardness requirement of between 33 and 43, inclusive, on the Rockwell C scale. Failure to meet this requirement shall be cause for rejection of the lifter and the lot it represents.

3.4 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.5 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

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4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

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ASTM E18: American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) If bid samples are required and the number of samples required.
(d) If first article samples are required, the number of samples required, and the subjective inspection requirements.
(e) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(f) If additional jaws are required (see 3.1.1).
(g) What preservation, packing, packaging, and marking of packages are required.

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the lifters specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	SIZE
5120-00-239-8686	Throat Depth: 9-3/4 to 10-1/4 Lifter Opens: 10 to 10-1/2 Valve Spring Diameter: 1-1/8 to 1-1/2

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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